

Scalability & Future Plan

Date	7 July 2026
Team ID	SWTID-2026-3702
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	1 Mark

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	Linear Regression model only — no ensemble methods	Limits prediction accuracy for non-linear patterns in HDI data	High
2	No database — predictions not stored or logged	Cannot track historical predictions or build analytics	Medium
3	Single-user local deployment — no cloud hosting	Application not accessible to external users or teams	High
4	Manual input only — no batch CSV prediction	Researchers cannot predict for multiple countries at once	Medium
5	No user authentication	Any user can access the application without identification	Low
6	Limited to 5 input features	Some HDI dimensions (education index, inequality) not included	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single-user localhost:5000	Deploy to Heroku/AWS EC2 with Gunicorn WSGI for multi-user support
2	Data Storage	Flat CSV file (195 rows)	Integrate PostgreSQL database for prediction logging and user data
3	Performance	Linear Regression (R-squared = 93.74%)	Implement Random Forest / XGBoost ensemble for improved accuracy
4	Security	No authentication	Add Flask-Login with JWT tokens for user authentication

5	Features	5 input features	Expand to include education index, inequality-adjusted HDI components
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Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Deploy to Heroku / AWS + Add user authentication	1 month after submission	Public accessibility, secure multi-user access
Phase 3	Implement Random Forest + XGBoost models for comparison	2 months	Improve R-squared from 93.74% to 96%+
Phase 4	Add batch CSV prediction upload and download results	2–3 months	Enable researchers to predict for 50+ countries at once
Phase 5	Integrate PostgreSQL for prediction history + user dashboard	3–4 months	Full production application with analytics
Phase 6	Add interactive Plotly/D3.js charts for visual HDI comparison	4–5 months	Rich data visualization for policy makers